Pramaan — Build & Demo Playbook

Oct 30, 2026 · @Akshay

Everything team CTRL + AI needs to build, demo and submit Pramaan for Forge '26. Submission closes Sun 1 Nov, 11:59 PM IST; finale demo Wed 4 Nov, 7:30 PM IST.

Pramaan in one page

Pitch: "Every week the government and regulators publish new rules in long English legal text. Pramaan reads each one live, tells you whether it applies to you, what to do and by when, in Telugu or Hindi, and proves every line with the exact sentence from the source."

Theme	Theme 1 — maps to the first listed problem, *Multilingual Policy, Rights & Scheme Explainer*, plus "complex regulatory compliance hurdles"
Users	(1) Small regulated businesses: NBFCs, co-operative banks, gold-loan firms, brokers, SEBI research analysts. (2) CAs and compliance consultants who track rules for many clients. (3) Citizens, through PIB scheme and policy announcements
Problem	Notifications are long, technical and English-only. Most don't apply to a given reader, but finding the one that does means reading all of them. Missing it means penalties or lost benefits
What it does	Live feed → "applies to you?" → checked action list with deadlines → regional-language voice explainer
USP	**Every claim is verified in code against the source text.** Anything the AI says that isn't in the document is rejected and shown as rejected. Chatbots and summarisers can't make that promise
Name / tagline	Pramaan (प्रमाण / ప్రమాణం, "proof") — "New rules, explained in your language, every line proved." Footer on every screen: independent tool, not affiliated with RBI, SEBI or the Government of India

Three words to say to judges: *live, verified, local-language.* Every demo moment should show one of them.

Where the 100 points come from

Live demo and robustness (30) plus AI guardrails and agentic execution (25) make up 55 points. Build for reliability and visible verification first, features second.

Pillar (points)	What judges look for	What Pramaan shows them
Live demo & robustness (30)	Runs live without breaking; handles bad input, missing data, network problems	Live fetch with timestamps; every failure path shows a clear message instead of crashing (Section 6); a "paste any circular link" box judges can try themselves
AI reasoning & agentic (25)	Plans, calls tools in several steps, handles errors, corrects itself; precise; few hallucinations; cites sources	A visible step log (fetch → read → extract → verify → retry → applicability → explain). The verifier rejects unsupported claims and the agent **re-extracts once** to fix them. Every line links to its source sentence
Relevance, impact & USP (25)	Real value, clear difference from chatbots and normal software, novel and scalable	Today's real circulars; profile-based "applies to you"; Telugu/Hindi voice; "verified or rejected" is the difference. Scales to any regulator with a feed
Technical depth (20)	Clean, modular code, low latency, more than a wrapper	Separate modules for sources, pipeline, verifier and LLM client; typed schemas; caching by document hash; parallel fetches; an eval script with measured precision

Feature priority, in order: live fetch → extraction → verifier → applicability → Telugu explainer + voice → self-correction retry → paste-a-link → deadlines (.ics). Then the three upgrades in the section "How much is enough": source side panel → weekly Telugu brief → Evidence page. Amendment diff only if ahead at hour 38.

Rules that can disqualify us

Each rule below is zero-tolerance. Treat this table as the team's contract.

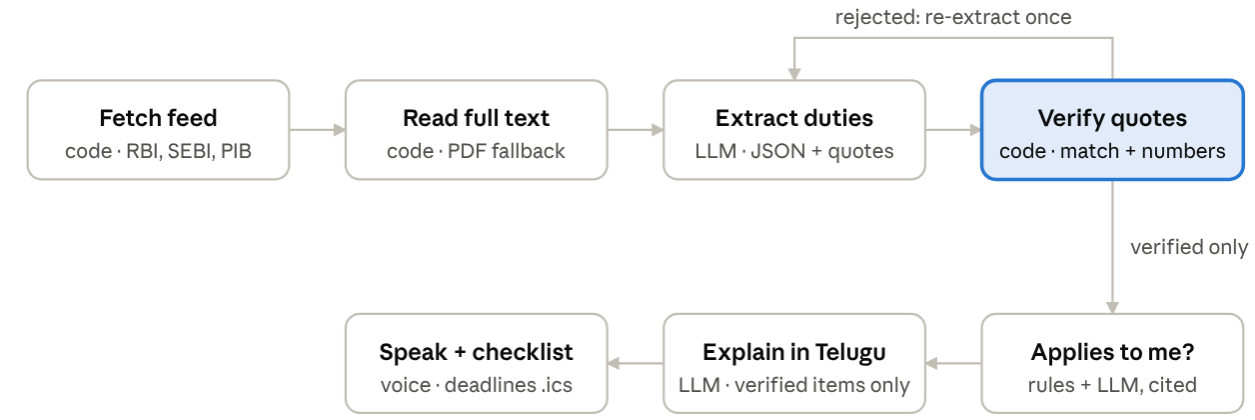
Rule	Do	Never
No dummy data	Fetch every circular live from RBI/SEBI/PIB at request time. Show source URL + fetch time on screen	Ship sample circulars, saved JSON outputs, or placeholder numbers in the repo or UI. No "demo mode" with canned answers
Built in the window	New repo created after 7:30 PM 30 Oct; commit small and often so history shows the build	Paste code from SAHVA, BOLVEDA, PlacementIQ or any earlier project. Libraries and starter templates are fine
Everyone understands it	Both of you can explain every module, prompt and data flow alone. Rehearse with each other	Let one person own a part the other can't explain
No fake live demo	Video and finale show the real deployed app working live	Edited screen recordings presented as live, or mockup screens
Fair play	Use Claude Code as your tool (allowed: it's a tool, like a library)	Code or help from another team or outsiders

Grey areas, handled safely:

- **Caching** fetched documents in memory for 30 minutes is fine: the data still came from a live source, and the UI shows when.
- **The evaluation set** (10 circulars you check by hand) lives in an `/eval` folder, is labelled as evaluation-only, and is never used by the app.
- **Typing a known circular link in the demo** is fine: it's user input, and the app still fetches and analyses it live.
- **The "test the verifier" button** (Section 7) must be labelled as a test that inserts one deliberately fake claim. Say so out loud.

How it works

The AI only reads and writes language; code does the fetching, checking and date handling. Nothing reaches the user unless its quote is found in the source.



Highlighted step is the guardrail. Rejected items are shown greyed, with the reason, never hidden.

Pramaan agent pipeline · 7 steps, 1 self-correction loop

The agent's tools

Tool	Type	What it does
`list_notices(source, days)`	code	Reads RBI / SEBI / PIB feeds; drops enforcement orders and recovery notices (SEBI: keep only `/legal/circulars/` links)
`fetch_document(url)`	code	Gets the page; strips HTML; keeps paragraph numbers; falls back to the PDF link (pdfplumber) if the page has no body text
`extract_obligations(text)`	LLM	Returns JSON only (schema below), temperature 0, long documents split by paragraph
`verify(obligations, text)`	code	The guardrail. Checks every quote, date and number against the source
`re_extract(rejected, text)`	LLM	One retry: "these items had no supporting quote; fix or drop them". Then verify again. Never more than one retry

Tool	Type	What it does
`check_applicability(profile, doc)`	rules + LLM	Rules first (who the circular is addressed to); LLM only when unclear; must cite a verified quote. Answers Applies / Doesn't apply / Needs review
`explain(verified, language)`	LLM	Writes the Telugu/Hindi explainer **from verified items only**, never from the raw document, so translation can't add claims
`speak(text, language)`	browser / TTS	Reads the explainer aloud

Obligation schema

```
{
  "who": "Research Analysts and Research Entities",
  "must_do": "Maintain call records of institutional clients",
  "deadline": "YYYY-MM-DD or null",
  "effective_from": "date or null",
  "paragraph": "4.2",
  "quote": "exact sentence copied from the circular",
  "penalty_or_consequence": "text or null"
}
```

The verifier, exactly

1. Normalise both texts: lower case, collapse spaces, unify quotes and dashes.
2. Exact substring match of the quote → **verified**.
3. Otherwise fuzzy match on a sliding window (rapidfuzz `partial_ratio` ≥ 92) → **verified (near match)**, shown with the matched source text.
4. Every date and every number in `must_do` / `deadline` must also appear in the quote → else **rejected: number not in source**.
5. No quote, or quote not found → **rejected**. Rejected items go to `re_extract` once, then are shown greyed with the reason.

Show the counts on screen: **"7 verified · 1 rejected · 1 fixed on retry"**. That line is your strongest score for guardrails and self-correction.

Data, stack and gotchas

Two deployments, no database: a custom **Next.js frontend on Vercel**, built with Claude Code, and a **FastAPI backend on Render** that runs the agent. The backend streams each agent step to the browser with Server-Sent Events, which is exactly the "real-time AI reasoning" the video requirement asks to see. LLM keys live only in the backend.

Sources

Source	Use	Gotcha
RBI notifications RSS	Main feed for banks, NBFCs, co-op banks	Shows only recent items (about a week in my 30 Oct test). Circular pages have full HTML text + a PDF link
SEBI RSS	Brokers, research analysts, listed firms	Mixed with enforcement and recovery orders: keep only `/legal/circulars/` links. Circular body may sit in an embedded PDF — check on hour 1
PIB RSS	Citizen-facing schemes and policy news	No dates in the feed. `Lang=1` returned Hindi in my test; find the English variant
Any RBI/SEBI/PIB link pasted by the user	Lets judges try their own	Allow only those domains; refuse others politely

Stack

Part	Choice	Why
Frontend	Next.js (React, TypeScript) + Tailwind + shadcn/ui, built with Claude Code; deployed on Vercel	A polished custom UI; judges see a real product, not a template
Backend	FastAPI (Python) on Render; Railway as the backup host	Agent pipeline in Python, where your AI work is strongest
Live step stream	Server-Sent Events: FastAPI `StreamingResponse` → browser `EventSource`; one event per tool call (name, status, time taken)	The step log that proves real agentic execution

Part	Choice	Why
Structure	Backend: `sources/`, `pipeline/`, `verify/`, `llm/`, `api/`, `eval/`. Frontend: `app/`, `components/`, `lib/api.ts`	Modularity is scored
Fetching	`httpx` with 10 s timeout, 2 retries, browser User-Agent, parallel fetches	Latency and robustness
Parsing	`feedparser`, `selectolax` or BeautifulSoup, `pdfplumber` fallback	
Verifier	`rapidfuzz`, `dateparser`, regex for numbers	
LLM	Gemini Flash (supports JSON-schema output; test its Telugu quality on hour 1); second provider as fallback behind one `llm.call()` function	Check today's free-tier limits before relying on them
Cache	In-memory TTL cache in the backend (`cachetools.TTLCache`, 30 min) keyed by URL + content hash	Fast repeat demos, no database, still live data
Profile	Browser `localStorage` (entity type, language), sent with each request	No login, no server storage
Voice	Browser `speechSynthesis`; a backend `/tts` endpoint using gTTS (unofficial Google Translate voice) as fallback	Telugu voice availability differs by device — test on the demo laptop
Deadlines	`.ics` file generated by the backend	
Security	Keys in Render/Vercel environment variables, never in the repo; CORS allows only your Vercel domain	

API contract (agree it at hour 4 so both halves build in parallel): `GET /notices?profile=&lang=` → list with applicability badges; `GET /analyze?url=&profile=&lang=` → SSE stream of steps, then the final result (obligations, verifier counts, explainer); `POST /verifier-test` → same, with one labelled fake claim; `GET /tts?text=&lang=` → MP3; `GET /health` → last fetch time per source.

Hour-1 tests (before building UI): deploy a bare FastAPI endpoint to Render and fetch all three feeds and two circular pages **from there**, not just your laptop. Government sites sometimes block cloud servers. If blocked, try Railway or Fly.io; note which host works. Also deploy an empty Next.js app to Vercel and confirm it can call the backend (CORS).

Edge cases the app must survive

The rubric names "unexpected inputs, missing data, unpredictable network". Build each row below and test it on purpose before recording.

Situation	What the app does
A source is down or times out	Shows "RBI unreachable, retried twice" on that source only; other sources still load
Cached copy exists and source is down	Shows the cached copy with "fetched 22 min ago"
Feed has no new items this week	Says so; offers "paste a circular link"
Item is an enforcement order or press note with no duties	Labels it "Informational, no obligations"; doesn't force the AI to invent some
Page has no body text	Falls back to its PDF; if that fails, says "couldn't read this document" with the link
Very long circular	Splits by paragraph, extracts per chunk, merges and removes duplicates
Document is in Hindi only	Extracts anyway; quotes stay in Hindi; explainer in the chosen language
LLM returns broken JSON	Retries once with the error message; then a second provider; then a clear error
LLM rate limit (429)	Waits and retries; switches provider; never a crash screen
All obligations rejected	Shows "Couldn't verify any obligations; read the original" with the link. Honest failure scores better than a confident wrong answer
User pastes a non-government URL, garbage text, or an empty box	Polite refusal naming the accepted sites
Profile is unclear for a circular	"Needs review" with the quote that made it unclear
Telugu voice missing on the device	Falls back to gTTS audio, or shows text only with a note

Situation	What the app does
Instructions hidden inside a document ("ignore previous instructions…")	The prompt treats document text as data; the verifier still blocks anything not quoted from the source

Wrap every external call in try/except with a message the user can understand. A judge pressing buttons quickly must never see a Python traceback.

Screens and demo script

Four screens are enough. Polish these instead of adding more.

1. **Home:** pick who you are (NBFC, Co-operative bank, Research Analyst, Stock broker, Citizen / small business) and a language (English, తెలుగు, हिंदी). Big **Check new rules** button. Results list: title, source, date, "fetched N min ago", and a badge — Applies / Doesn't apply / Needs review.
2. **Circular view:** live step log at the top (each tool call with a tick and time taken); action cards (who, what, by when) each with its highlighted source sentence and paragraph link; the verifier bar "7 verified · 1 rejected · 1 fixed on retry"; rejected items greyed with the reason; Telugu explainer with a play button; **Add deadlines to calendar** (.ics).
3. **Paste a link:** any RBI / SEBI / PIB URL → the same analysis.
4. **How it works:** the pipeline diagram, the eval result, and a **Test the verifier** button that adds one clearly labelled fake obligation (e.g. "pay a ₹10 lakh fee by 1 Dec") so everyone sees it rejected live.

Demo video (2:30, recorded from the live deployed app)

Time	Show	Say
0:00–0:20	RBI's notifications page scrolling	"RBI, SEBI and the government publish new rules every week, in long English legal text. A small lender or accountant has to read all of them to find the one that matters."
0:20–0:45	Pick "Research Analyst", Telugu; press Check new rules	"Pramaan pulls this week's notices live — here's the fetch time — and sorts them by whether they apply to you."

Time	Show	Say
0:45–1:35	Open the one that applies; step log runs; action cards appear	"The agent reads the full circular, extracts each duty with the exact sentence, then code checks every quote, date and number. Seven verified, one rejected, one fixed on retry."
1:35–1:55	Press play on the Telugu explainer	Let it play 10 seconds
1:55–2:15	Switch profile to Co-operative bank	"Different reader, different answer — the SEBI rule drops out, an RBI one lights up."
2:15–2:30	How it works + eval number	"Live data, verified answers, local language. That's Pramaan."

Finale live demo (4 Nov, online)

The format and length aren't published; prepare a 3-minute run and a 5-minute run. Same flow as the video, plus two moves that only work live:

- **Ask a judge to pick** any circular from the list, or paste a link of their choice. Confidence that it works on anything is the robustness score.
- **Press Test the verifier** and say clearly: "This button inserts one fake claim on purpose, to show you the checker catches it."

How much is enough to make judges say "wow"

One flawless flow beats ten shaky features: robustness alone is 30 points. Enough is the MVP plus three upgrades, all working every time.

What makes it interesting (not the tools you used): judges never see Claude Code. A plain "PDF summary" of a dry topic is forgettable; four moments make Pramaan memorable — the agent's steps streaming live, a fake claim visibly caught, Telugu voice, and a judge choosing the input.

Build order and the wow each adds

Order	Feature	Wow	Effort	Decision
MVP	Live fetch, extraction, verifier + one retry, applicability, Telugu/Hindi voice, paste-a-link, graceful errors	Core	—	Must be done by hour 30
1	**Source side panel:** click a duty, the original scrolls to its highlighted sentence	High	Low	Build
2	**"This week for you":** one 60-second Telugu audio brief of every notice that applies to the profile	High	Low (reuses parts)	Build; also your social-media clip
3	**Evidence page:** measured accuracy on 10 hand-checked circulars, time and cost per circular	Medium	Low	Build; answers "how do you know?" with numbers
4	**"What changed" diff** for amendment circulars	High	Medium	Only if ahead at hour 38
—	Asking questions by voice in Telugu	High	High, risky live	Skip
—	WhatsApp delivery, login, alerts, more regulators, mobile app	Low in a demo	High	Skip

Working with Claude Code

- Give it this playbook first, then **one feature per session** with a short spec: inputs, outputs, error cases.
- Ask for pytest tests on the verifier and parsers; run them before every deploy.
- Commit after every working step; tag `mvp` at hour 30 and deploy it immediately.
- Hide unfinished features behind a switch so the live app never shows half-built screens.
- **Feature freeze at hour 44 (Sun 1 Nov, 3:30 PM IST).** After that: fixes, video, README, deck only.

- Write the prompts and verifier rules yourselves, read every module it writes, and explain each other's parts aloud. The rules require that both of you can explain the code without help.
- "Unauthorized assistance" almost certainly means other people, and this is a GenAI event; if you want certainty, ask the organisers in one line.

Other things to keep in mind

- **The 4 Nov finale will show different circulars from your video.** That proves it's live. Run the checklist that morning and keep 2–3 fresh links ready to paste.
- **Keep the app and keys alive until 5 Nov:** don't rotate keys, watch quotas, wake the free host before the demo.
- **Submit by Sun 1 Nov, 9:30 PM IST,** not 11:59. Uploads and links fail at the last minute.
- **Sleep in shifts.** A tired team breaks the build on Sunday evening.
- **Social posts after submitting,** asking for reshares (5 points each).
- **Branding:** no RBI/SEBI names or national emblem in the logo; keep the "independent tool" footer.

Before every demo, and judge questions

The numbers in the script above are examples: say whatever the live run actually shows. Never script a result.

Checklist (run 30 minutes before the video and before the finale)

- [ ] Open the Vercel app and hit the backend /health to wake Render (free hosts sleep when idle); load once fully
- [ ] Press Check new rules for each profile; confirm each source loads and note which circulars give a good result **today**
- [ ] Keep 2–3 real circular links in a notes file as your paste-a-link backups
- [ ] Check LLM quota and both API keys; confirm the fallback provider works
- [ ] Play the Telugu and Hindi voice on the exact laptop and browser you'll present from
- [ ] Wired internet or strong Wi-Fi, with a phone hotspot ready
- [ ] Close other tabs and notifications; browser zoom 110–125% so text is readable on a shared screen
- [ ] Backup backend on Railway deployed with the same code and keys; a frontend environment switch points to it in one step
- [ ] Both teammates have run the full demo alone at least once

Questions judges are likely to ask

Question	Answer
How do you stop hallucinations?	Every obligation needs a verbatim quote; code checks the quote, dates and numbers against the source; anything unmatched is rejected and shown. The explainer is generated only from verified items
What if the AI misses an obligation?	Recall is the harder problem. We measured it on 10 hand-checked circulars: [your numbers]. The original is always one tap away, and "Needs review" is used when unsure
Is this just RAG plus a summary?	No: it's an agent with tools — fetch, parse, extract, verify, retry, classify, translate — and a code verifier that can overrule the model
Where is the data from? Is it live?	Official RBI, SEBI and PIB feeds, fetched at request time; the fetch time is on screen. Nothing is stored or pre-written
Why no database?	Every answer is computed live from the source; an in-memory 30-minute cache keeps it fast. A database comes later for alerts and history
Who pays for this?	CA and compliance firms tracking many clients, and small regulated lenders; a free tier for citizens on PIB schemes
How does it scale?	Any regulator with a feed or listing page is a new source module: GST/CBIC, DGFT, state gazettes
What did you build in 48 hours vs reuse?	Libraries only (Next.js, FastAPI, httpx, rapidfuzz, LLM SDKs). Every prompt, tool, verifier rule and UI was written in the window; the commit history shows it
Why is Telugu output trustworthy?	It's produced only from verified English items, and each Telugu point links back to its source sentence

Submission pack

All six items are mandatory; finish the text and README by Sunday 6 PM so the last hours go to the video and deploy checks.

Submission text (draft — edit with your real numbers)

Problem statement — Target issue & context: Indian regulators and ministries publish new rules every week as long English legal documents. Small regulated businesses and the accountants who serve them must read each one to find out whether it applies, what it requires and by when. Missed obligations lead to penalties; citizens miss scheme deadlines for the same reason.

Target audience: small NBFCs, co-operative banks, brokers and research analysts; CA and compliance firms; citizens and small businesses affected by government schemes.

Solution — high-level overview: Pramaan is an agent that fetches new RBI, SEBI and PIB notifications live, reads the full text, extracts each obligation with its exact source sentence, and verifies every quote, date and number in code before showing it. It checks whether each notice applies to the user's profile and explains the verified duties in Telugu or Hindi, with voice.

Technical architecture: Python + FastAPI backend on Render, Next.js frontend on Vercel, agent steps streamed live over Server-Sent Events; tool-based agent pipeline (fetch, parse, extract, verify, re-extract, classify, explain); Gemini for extraction and translation with a fallback provider; rapidfuzz + dateparser verifier; live official feeds; in-memory cache, no stored data.

README must include

- One-paragraph pitch and the live app link
- Architecture diagram (export the pipeline picture above)
- Data sources with links, and a line that all data is fetched live
- Setup: `backend — pip install -r requirements.txt, copy .env.example to .env and add keys, uvicorn api.main:app; frontend — npm install, set NEXT_PUBLIC_API_URL, npm run dev`
- How the verifier works, and the eval method + results
- Known limits, and team roles

Deck (6 slides, following the organisers' structure)

1. **Title & team** — Pramaan, tagline, CTRL + AI, NIT Warangal
2. **Problem** — "a week of RBI notices" screenshot; who suffers and how

3. **Solution & features** — one screenshot of the circular view with its verifier bar
4. **Architecture & innovation** — the pipeline diagram; "code can overrule the model"
5. **Results** — eval numbers, latency per circular, sources covered
6. **Roadmap & demo summary** — more regulators, WhatsApp delivery, alerts; demo link + QR

Audience Favourite (separate from the judges' prizes)

Scoring: reshare 5 points, comment 3, like 1, added across LinkedIn, Instagram and X; **one post per platform**; open until **4 Nov, 11:59 PM IST**. Reshares are worth five likes, so ask for reshares.

- Post all three on Sunday night or Monday morning, after submitting
- LinkedIn: 45-second screen clip of the Telugu voice moment + one line on the verifier
- Instagram: Reel, same clip, captions in Telugu and English
- X: the clip + "RBI changed N rules this week. Here's who they hit."
- Send to NIT Warangal batch groups, clubs and family with a direct ask to reshare and comment
- Don't let this cut into build time before Sunday's deadline

Not verified yet — check in the first two hours

These could change the build plan; each takes minutes to test.

- [ ] RBI, SEBI and PIB fetches work **from the Render backend** (tested only from my side on 30 Oct)
- [ ] Whether SEBI circular pages carry body text or only an embedded PDF
- [ ] The English variant of the PIB feed (`Lang` parameter)
- [ ] Current free-tier limits of the LLM provider(s) you choose
- [ ] Telugu and Hindi voices in your demo browser; gTTS as fallback
- [ ] Render free-tier sleep behaviour and cold-start time
- [ ] The finale format and time per team (4 Nov) — ask the organisers

Verified on 30 Oct: RBI notifications RSS (items 23–27 Oct 2026), an RBI circular page with full HTML text and PDF link, SEBI RSS (circulars of 22–23 Oct), PIB RSS. Strategy, scores and the pitch are my judgement, not data.